

Overview

SkillBridge - Java Basics

What is Java?

Java is a popular and powerful programming language, created in 1995. It is owned by Oracle, and more than 3 billion devices run Java.

- Mobile applications (specially Android apps)
- Desktop applications
- Web applications
- Web servers and application servers
- Games
- Database connection
- And much, much more!

Features of Java

Simple and Easy to Learn

Object-Oriented

Platform Independent

Secure

Robust

Multithreaded

High Performance

Java works on 3 main components:

- JDK (Java Development Kit) – Used to develop Java programs
- JRE (Java Runtime Environment) – Used to run Java programs
- JVM (Java Virtual Machine) – Converts bytecode into machine code

Why Use Java?

Java works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc.)

It is one of the most popular programming languages in the world

It has a large demand in the current job market

It is easy to learn and simple to use

It is open-source and free

It is secure, fast and powerful.

- What is java syntax and **main()** method?

Java syntax means the set of rules that define how to write programs correctly — just like grammar rules in English. If the syntax is wrong, the program will show an **error**.

The **main()** method is required in every Java program.

```
public static void main(String[] args)
```

What is java comment?

Comments can be used to explain Java code, and to make it more readable. It can also be used to prevent execution when testing alternative code

Two types of comment

- Single line comment : Single-line comments start with two forward slashes (//)
- Multiline comments : Multi-line comments start with `/*` and ends with `*/`.

What are Java variable types?

Variables are containers for storing data values.

- **String** - stores text, such as "Hello". String values are surrounded by double quotes

```
String name = "John"      System.out.println(name);
```

- **int** - stores integers (whole numbers), without decimals, such as 123

```
int myNum = 15            System.out.println(myNum);
```

- **float** - stores floating point numbers, with decimals, such as 19.9

```
float myFloatNum = 5.99f;
```

- **char** - stores single characters, such as 'a'. Char values are surrounded by single quotes

```
char myLetter = 'D';
```

- **boolean** - stores values with two states: true or false

```
boolean myBool = true;
```

What is Java data types?

- Primitive data types - includes **byte**, **short**, **int**, **long**, **float**, **double**, **boolean** and **char**
- Non-primitive data types - such as **String**, **Arrays** and **Classes** (you will learn more about these in a later chapter)

What is a java operator?

Java operators are special symbols that perform specific operations on one, two, or three operands, and then return a result. They are the building blocks for creating expressions in Java programs.

Types of java operator

- 1) **Arithmetic Operators:** Used for basic mathematical operations.
 - a) Examples: `+` (addition), `-` (subtraction), `*` (multiplication), `/` (division), `%` (modulus).
- 2) **Relational Operators:** Used to compare two values.
 - a) Examples: `==` (equal to), `!=` (not equal to), `>` (greater than), `<` (less than), `>=` (greater than or equal to), `<=` (less than or equal to).
- 3) **Logical Operators:** Used to combine multiple boolean expressions.
 - a) Examples: `&` (logical AND), `||` (logical OR), `!` (logical NOT).
- 4) **Bitwise Operators:** Used for operations on bits.
 - a) Examples: `&` (bitwise AND), `|` (bitwise OR), `^` (bitwise XOR), `~` (bitwise NOT), `<<` (left shift), `>>` (right shift).
- 5) **Assignment Operators:** Used to assign values to variables.
 - a) Examples: `=` (simple assignment), `+=` (add and assign), `-=` (subtract and assign).
- 6) **Unary Operators:** Operate on a single operand.
 - a) Examples: `++` (increment), `--` (decrement), `-` (negation).
- 7) **Ternary Operator:** A shorthand for the `if-else` statement.
 - a) Syntax: `condition ? expression1 : expression2`.
- 8) **Instanceof Operator:** Checks whether an object is an instance of a specific class or subclass.*

Java Loops: In Java, loops are control flow statements that allow you to execute a block of code repeatedly based on a specified condition. There are several types of loops in Java:

1. **For Loop:** A for loop is used when you know in advance how many times you want to execute a statement or a block of statements.

```
java
for (int i = 0; i < 10; i++) {    System.out.println(i);}
```

2. **While Loop:** A while loop continues to execute as long as a specified condition is true. It checks the condition before the loop's body is executed.

```
java
int i = 0; while (i < 10) { System.out.println(i); i++; }
```

3. **Do-While Loop:** A do-while loop is similar to a while loop, but it checks the condition after the loop's body has been executed. This ensures that the loop's body is executed at least once.

```
java
int i = 0; do { System.out.println(i); i++; } while (i < 10);
```

4. **Enhanced For Loop (For-Each Loop):** This loop is used to iterate over arrays and collections without using an index variable. It's more convenient for traversing elements.

```
java
int[] numbers = {1, 2, 3, 4, 5}; for (int number : numbers) { System.out.println(number); }
```

- Java Algorithms: A **Java algorithm** means a **step-by-step procedure written in Java to solve a problem**.
- 👉 An **algorithm** is the logic or steps to solve a problem.
- 👉 **Java** is the programming language used to implement that logic.
- So, a Java algorithm = **problem-solving steps written using Java syntax**.

Background

BACKGROUND

History with the company

In a paragraph or less, provide an overview of the history with this company. This could include interactions, key events, communications, and milestones.

Key decision-makers

Name	Title	POC	Contact information	Notes
 Person	Title	 Person	Add information	Add notes
 Person	Title	 Person	Add information	Add notes
 Person	Title	 Person	Add information	Add notes

Current product and services

- List the types of products and services this customer prefers
- Preference 2
- Preference 3

Account strategy

ACCOUNT STRATEGY

Indicate how the account strategy plan aligns with the overall goals of the account. Explain how the strategy will be used to pursue specific initiatives to help achieve company goals.

ACTIONS	RESOURCES	METRICS	DUE DATE	OWNER
Add action	Add resources	- Metric 1 - Metric 2	Date	Person Person
Add action	Add resources	- Metric 1 - Metric 2	Date	Person Person
Add action	Add resources	- Metric 1 - Metric 2	Date	Person Person

ACTIONS	RESOURCES	METRICS	DUE DATE	OWNER
Add action	Add resources	- Metric 1 - Metric 2	📅 Date	👤 Person 👤 Person
Add action	Add resources	- Metric 1 - Metric 2	📅 Date	👤 Person 👤 Person

Competitive and SWOT analysis

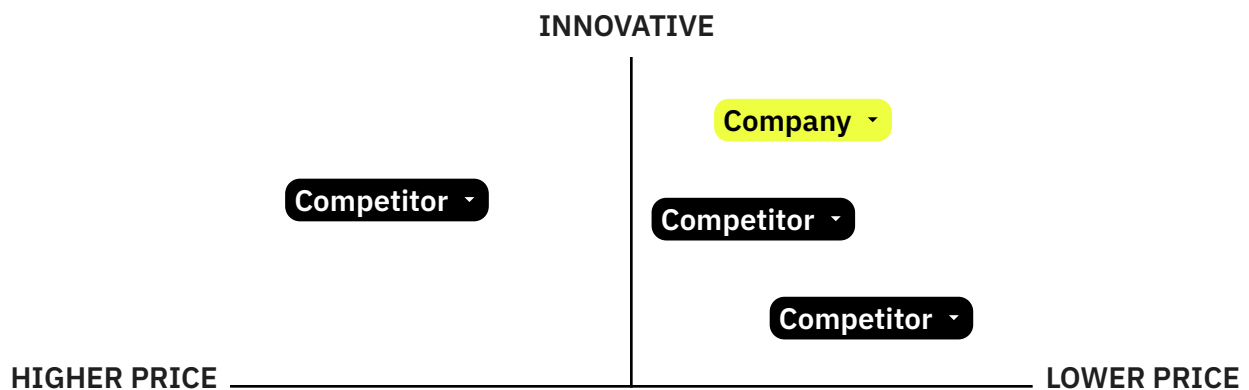
COMPETITIVE ANALYSIS

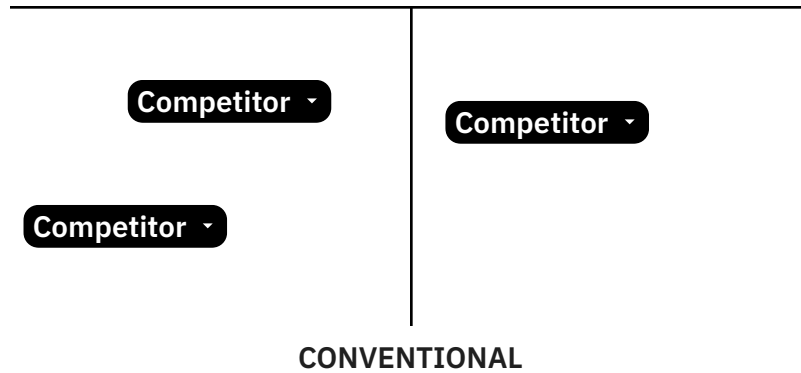
Describe their competitors

- List companies offering similar products or services. Be specific and include large and small competitors.
- Consider alternative products or services that might meet the same customer needs

Summarize their competitive advantage

- Clearly articulate how your client differentiates itself from the competition
- Highlight market opportunities
- Support the competitive advantages with data whenever possible





SWOT ANALYSIS

Strengths	Weaknesses
Internal factors that give the account a competitive advantage.	Internal factors that can hinder the account's progress.
Focus on unique capabilities, resources, or market position.	Identify areas for improvement or potential vulnerabilities.
Examples: Strong brand reputation, innovative products, loyal customer base, efficient operations, skilled workforce.	Examples: Limited market share, outdated technology, high costs, lack of skilled personnel, weak distribution network.

Opportunities

External factors that the account can use for growth or improvement.

Identify emerging trends, market gaps, or favorable conditions.

Examples: New market segments, technological advancements, changing regulations, economic growth, competitor weaknesses.

Threats

External factors that can negatively impact the account.

Identify potential risks and challenges.

Examples: Intense competition, economic downturn, changing customer preferences, disruptive technologies, regulatory changes.

Interaction log

INTERACTION LOG

Date	POC	Type of interaction	Notes	Next steps
Date	Person	Meeting ▾	Interested in new product	Ongoing ▾
Date	Person	Phone call ▾	Notes	Resolved ▾
Date	Person	Email ▾	Notes	Escalated ▾

ACTION ITEMS

Action item	Assigned to	Deadline	Status
Send customer new catalog	Person	Date	Completed ▾
Action item 2	Person	Date	Completed ▾
Action item 3	Person	Date	Completed ▾

MEETING NOTES

To add a meeting notes building block, type “@meeting notes”. Search for or select a meeting to add attendees, notes, and action items.